Soft Computing
Introduction

Computing

- ***The process or act of calculation***
- **The action of mathematical calculation.**
- Computing is ***any activity that uses computers*** to manage, process, and communicate information.
- It includes development of both ***hardware and software***.
- Computing is a ***critical, integral component of modern industrial technology***.
- Major computing disciplines include computer engineering, software engineering, computer science, information systems, and information technology.

Brain VS Computer

	Brain	Computer
Processing Elements	10^{10} neurons	10^{8} transistors
Element Size	10^{-6} m	10^{-6} m
Energy Use	30 W	30 W (CPU)
Processing Speed	10^{2} Hz	10^{12} Hz
Style Of Computation	Parallel, Distributed	Serial, Centralized
Energetic Efficiency	10^{-16} joules/opn/sec	10^{-6} joules/opn/sec
Fault Tolerant	Yes	No
Learns	Yes	A little

Soft Computing

- Soft computing is the use of **approximate calculations** to provide imprecise but usable solutions to complex computational problems.
- The approach enables solutions for problems that may be **either unsolvable or just too time-consuming to solve** with current hardware.
- Soft computing is sometimes referred to as **computational intelligence.**
- With ***the human mind as a role model***, soft computing is tolerant of partial truths, uncertainty, imprecision and approximation, unlike traditional computing models.
- The tolerance of soft computing allows researchers to approach some problems that ***traditional computing can't process***.

Continue...

- Soft computing differs from conventional (hard) computing in that, unlike hard computing, ***it is tolerant of imprecision, uncertainty, partial truth, and approximation.***
- ***In effect, the role model for soft computing is the human mind.***
- It ***does not require any mathematical modelling*** for solving any given problem
- It ***gives different solutions when we solve a problem of one input from time to time***
- ***Uses some biologically inspired methodologies*** such as genetics, evolution, particles swarming, the human nervous system, etc.
- ***Adaptive in nature.***

Few Facts on Soft Computing

- ***tolerance of imprecision:*** the result obtained using soft-computing is not precise.
- **uncertainty:** the soft-computing algorithm may give different results every time for the same problem.
- **robustness:** soft-computing algorithms can tackle any kind of input noise
- **low solution cost:** soft-computing makes it feasible to solve some of the problems which could be computationally very expensive if solved using hard computing.

Advantages of Soft Computing

- Since Soft computing methods ***do not call for wide-ranging mathematical formulation*** pertaining to the problem, the need for explicit knowledge in a particular domain can be reduced.
- These tools can handle ***multiple variables*** simultaneously.
- For optimization problems, the solutions ***can be prevented from falling into local minima*** by using global optimization strategies.
- These techniques are ***mostly cost effective***.
- **Dependency on expensive traditional simulations packages can be reduced to some degree** by efficient hybridization of soft computing methods.
- These methods are generally ***adaptive in nature and are scalable***.

Applications of Soft Computing

- Image processing
- Data Compression
- Fuzzy Logic Control
- ***Automative systems and Manufacturing***
- Neuro-fuzzy systems
- ***Decision-support systems***
- ***System Control***
- ***Prediction***

and many more.

Hard Computing

- Hard computing, i.e., ***conventional computing, requires a precisely stated analytical model*** and often a lot of computation time.
- Many ***analytical models*** are valid for ideal cases.
- ***Real world problems exist in a non-ideal environment***.
- Premises and guiding principles of Hard Computing are – ***Precision, Certainty, and rigor***.
- ***Many contemporary problems do not lend themselves to precise solutions*** such as – Recognition problems (handwriting, speech, objects, images – Mobile robot coordination, forecasting, combinatorial problems etc.

Difference Between Hard Computing and Soft Computing

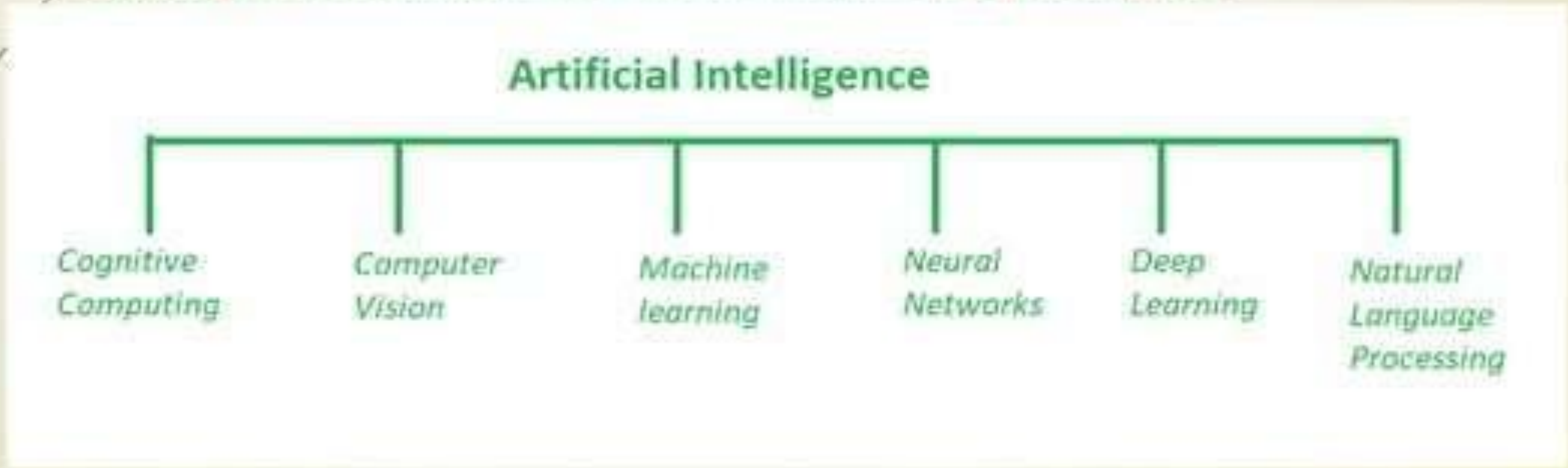
Hard Computing	Soft Computing
The analytical model required by hard computing must be precisely represented	It is based on uncertainty, partial truth tolerant of imprecision and approximation.
Computation time is more	Computation time is less
It depends on binary logic, numerical systems, crisp software.	Based on approximation and dispositional.
Hard computing performs sequential computations.	Soft computing can perform parallel computations.
Hard computing works on exact data.	Soft computing works on ambiguous and noisy data.

Difference Between Hard Computing and Soft Computing....

Hard Computing	Soft Computing
Hard computing uses two-valued logic.	Soft computing will use multivalued logic.
Hard computing is settled.	Soft computing incorporates randomness .
Hard computing requires programs to be written.	Soft computing will emerge its own programs.
Hard computing produces precise results.	Soft computing produces approximate results.
Hard computing is deterministic in nature.	Soft computing is stochastic in nature.

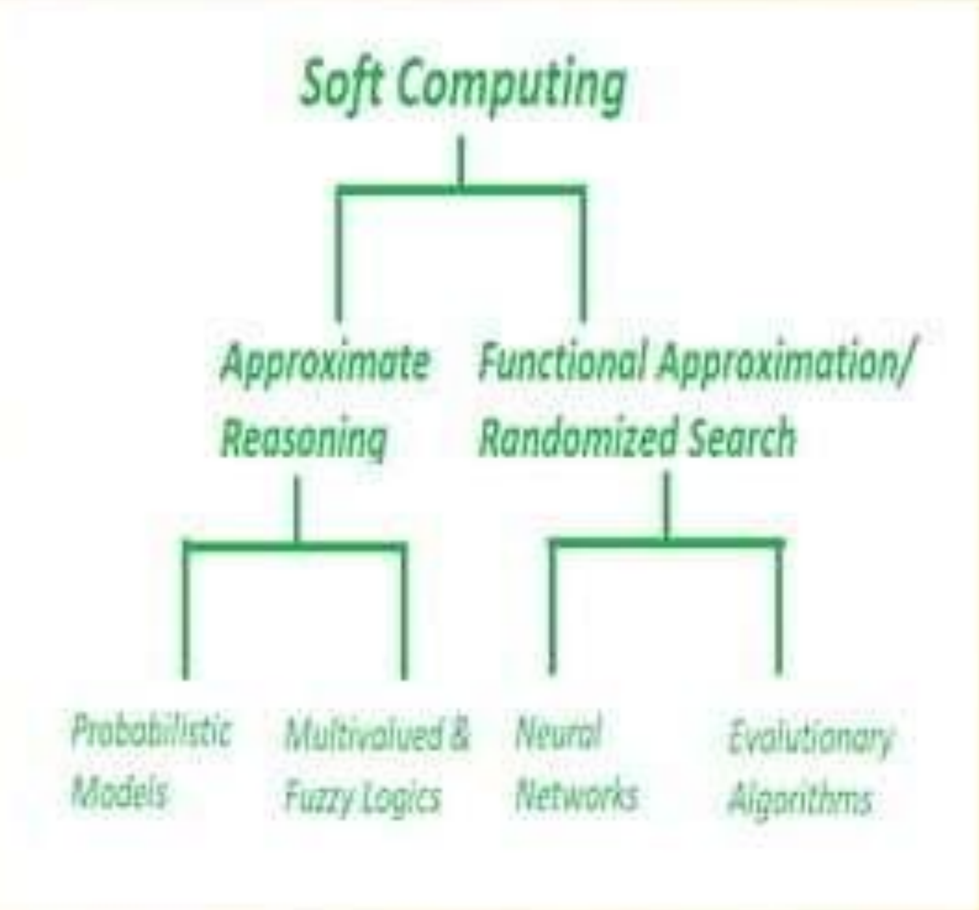
Artificial Intelligence (AI)

- ***AI manages more comprehensive issues of automating a system.*** This computerization should be possible by utilizing any field such as image processing, cognitive science, neural systems, machine learning etc.
- AI manages the making of machines, frameworks and different gadgets savvy by ***enabling them to think and do errands as all people generally do.***



Soft Computing

- Soft Computing could be a ***computing model*** evolved to resolve the non-linear issues that involve unsure, imprecise and approximate solutions of a tangle.
- These sorts of issues square measure thought of as real-life issues wherever the ***human-like intelligence is needed to resolve it.***



Differentiate Between AI and Soft Computing

AI	Soft Computing
Artificial Intelligence is the art and science of ***developing intelligent machines.***	Soft Computing ***aims to exploit tolerance for uncertainty, imprecision, and partial truth***
AI plays a fundamental role in ***finding missing pieces between the interesting real world problems.***	Soft Computing comprises techniques which are ***inspired by human reasoning*** and have the potential in handling imprecision, uncertainty and partial truth.
Branches of AI : 1. Reasoning 2. Perception 3. Natural language processing	**Branches of soft computing :** 1. Fuzzy systems 2. Evolutionary computation 3. Artificial neural computing

Differentiate Between AI and Soft Computing

AI	Soft Computing
AI has ***countless appl***ications in healthcare and widely used in analyzing complicated medical data.	They are used in science and engineering disciplines such as ***data mining, electronics, automotive, etc.***
Goal is to stimulate human-level intelligence in machines	***It aims at accommodation with the pervasive imprecision of the real world.***
They require programs to be written.	They not require all programs to be written, they can evolve its own programs.
They require exact input sample.	They **can deal with ambiguous and noisy data.**

Artificial Neural Network

Neural Network is a network of artificial neurons, ***inspired by biological network of neurons***, that uses mathematical models as information processing units to discover patterns in data which is too complex to notice by human.

There are ***millions of neurons*** in the human brain, and the ***information passes from one neuron to another***. A neural network works similar to that and is capable of performing computations faster.

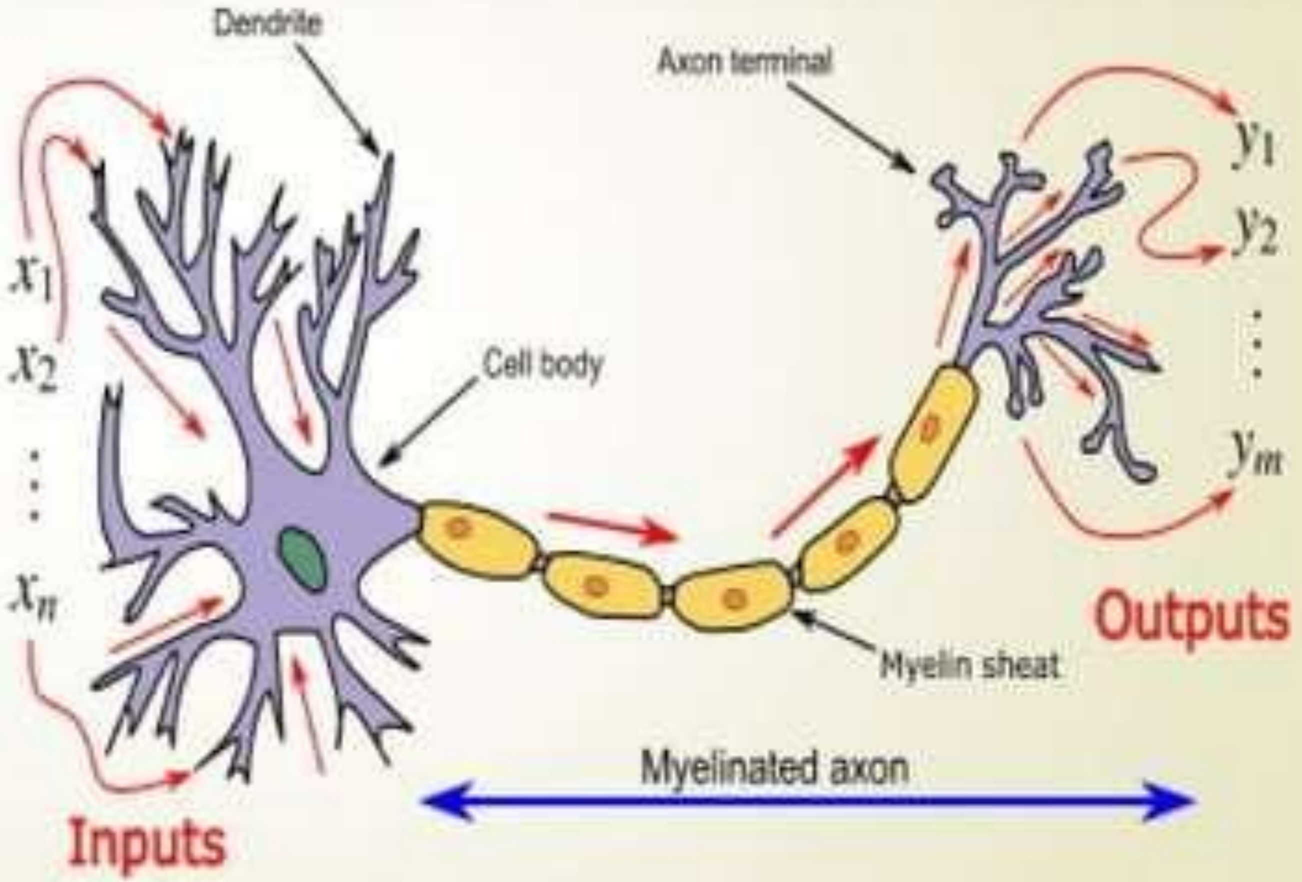
Artificial Neural Network

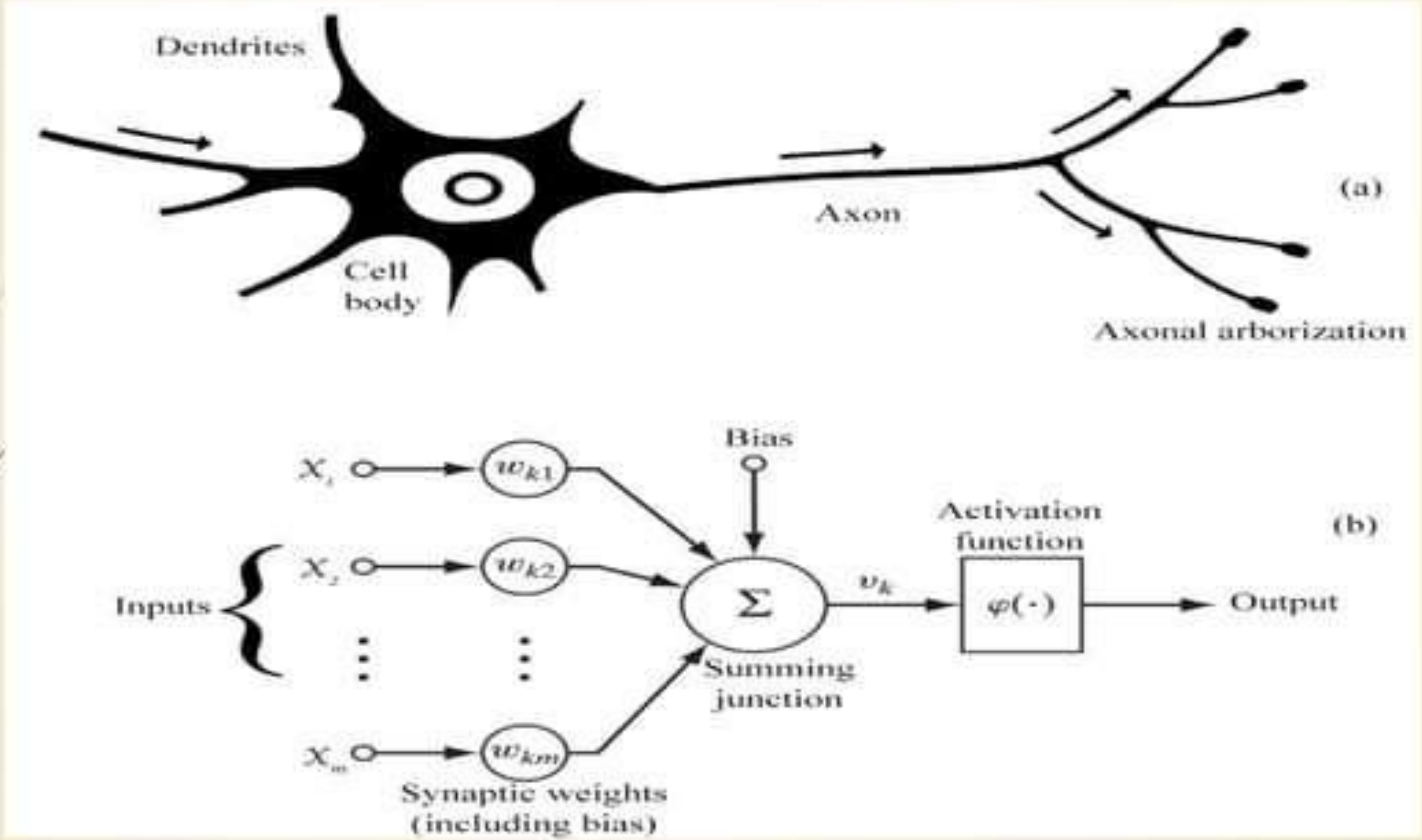
Dendrite: Receives signals from neighbouring neurons

Soma: Accumulates the signals received through the dendrites

Axon: Transmits signal from soma to the axon terminals

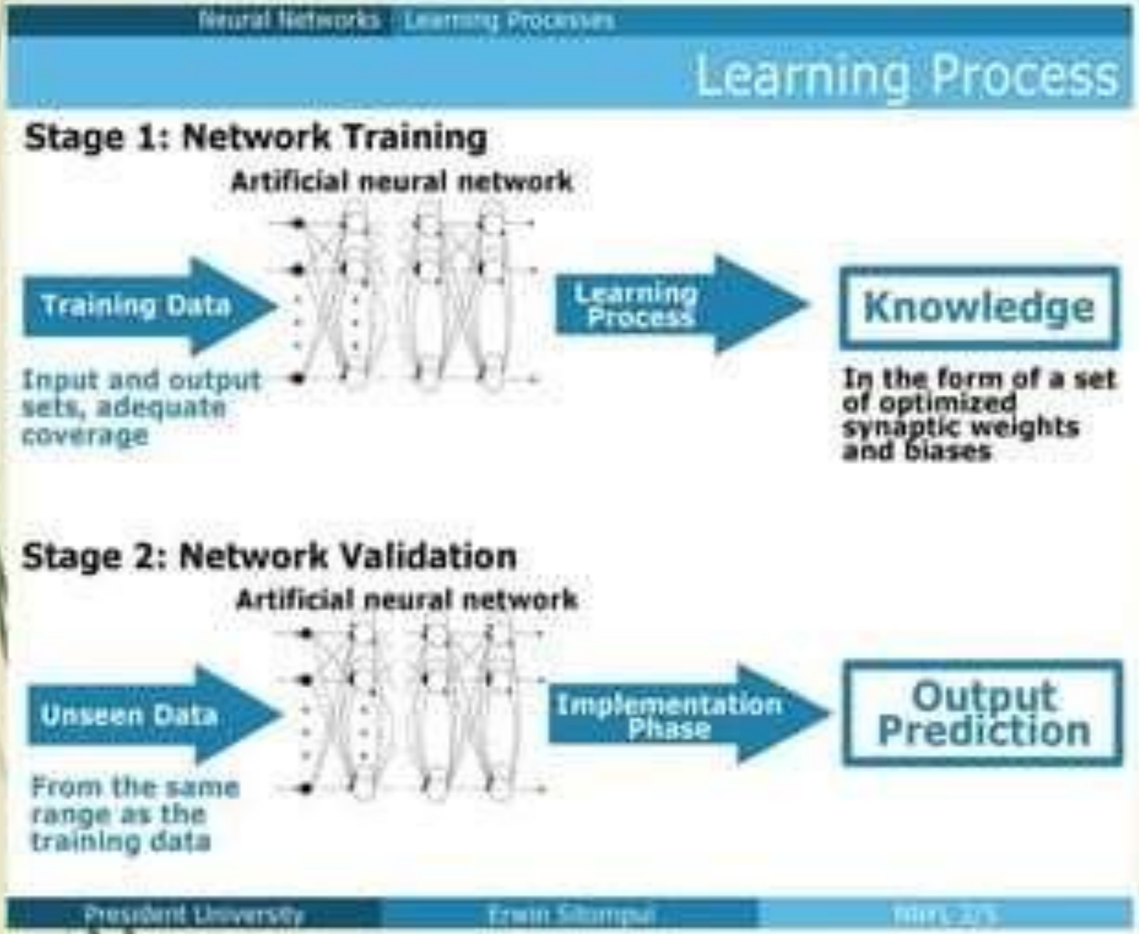
Axon terminals: propagates stimulus to neighbouring neurons

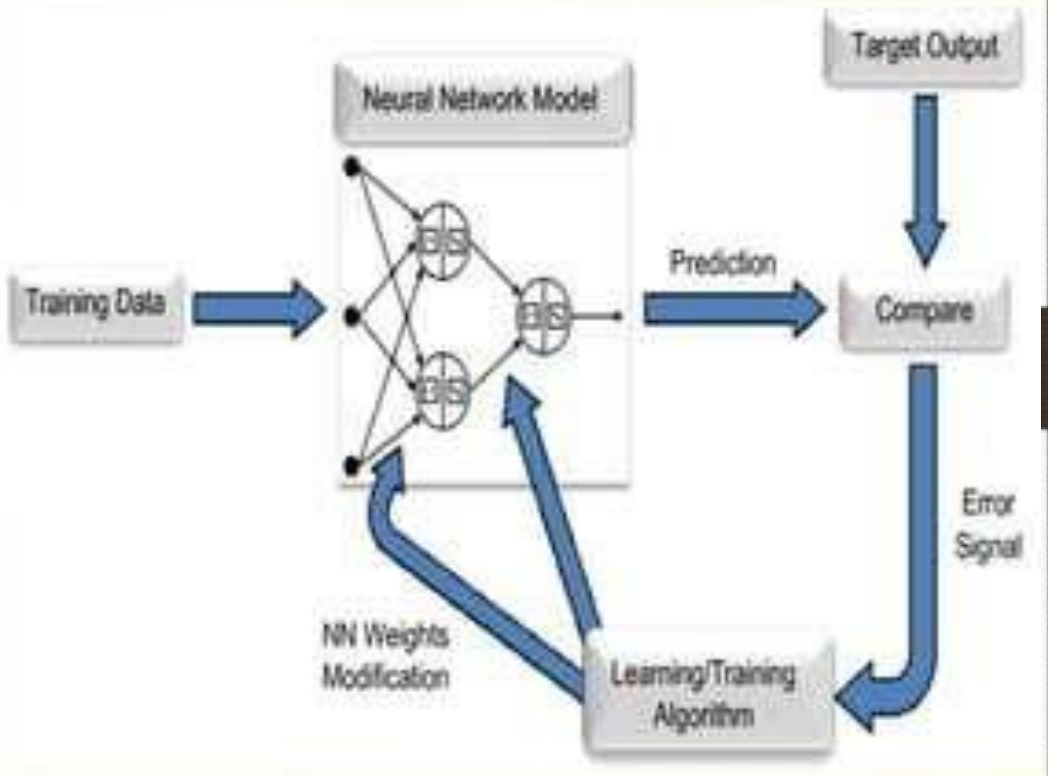


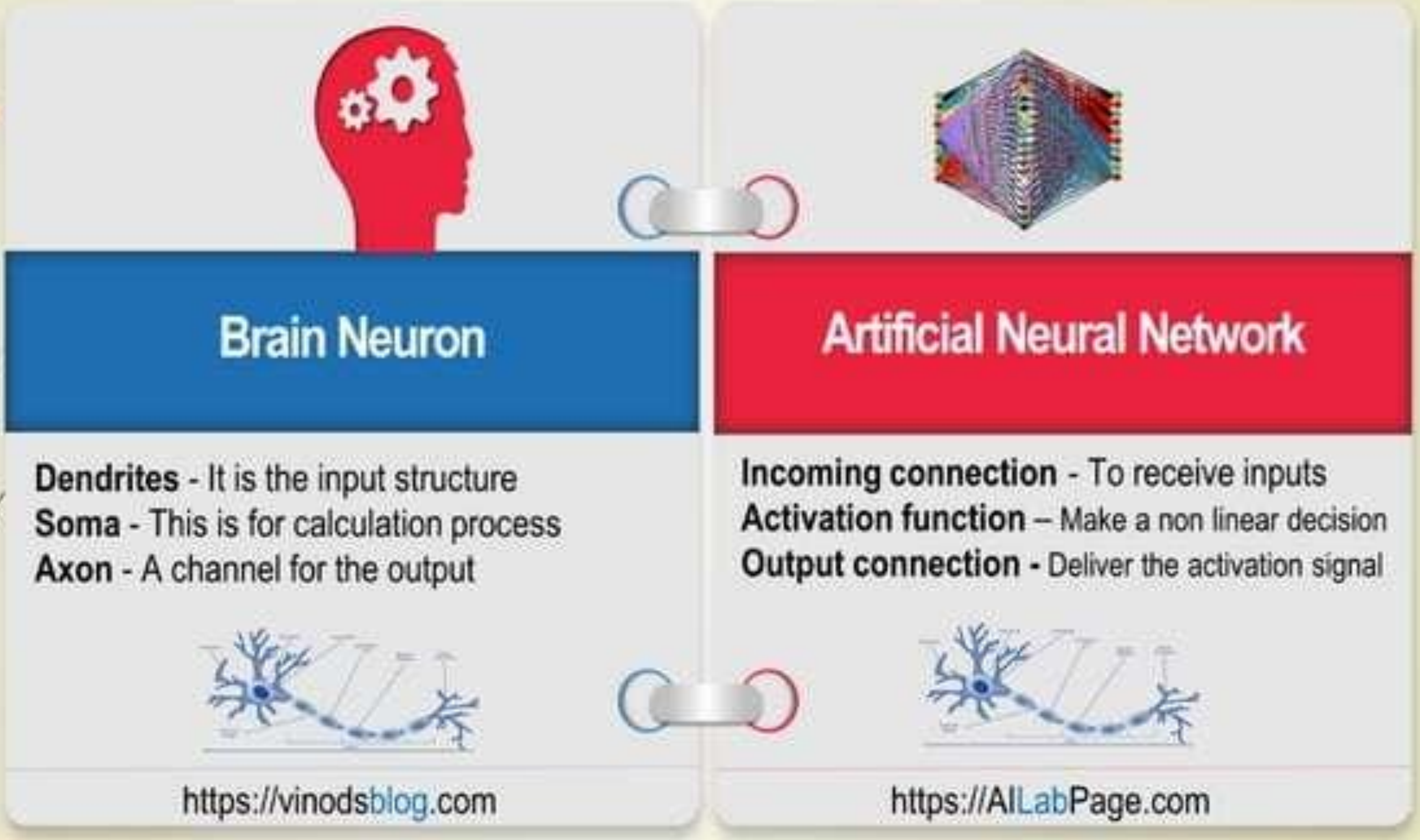


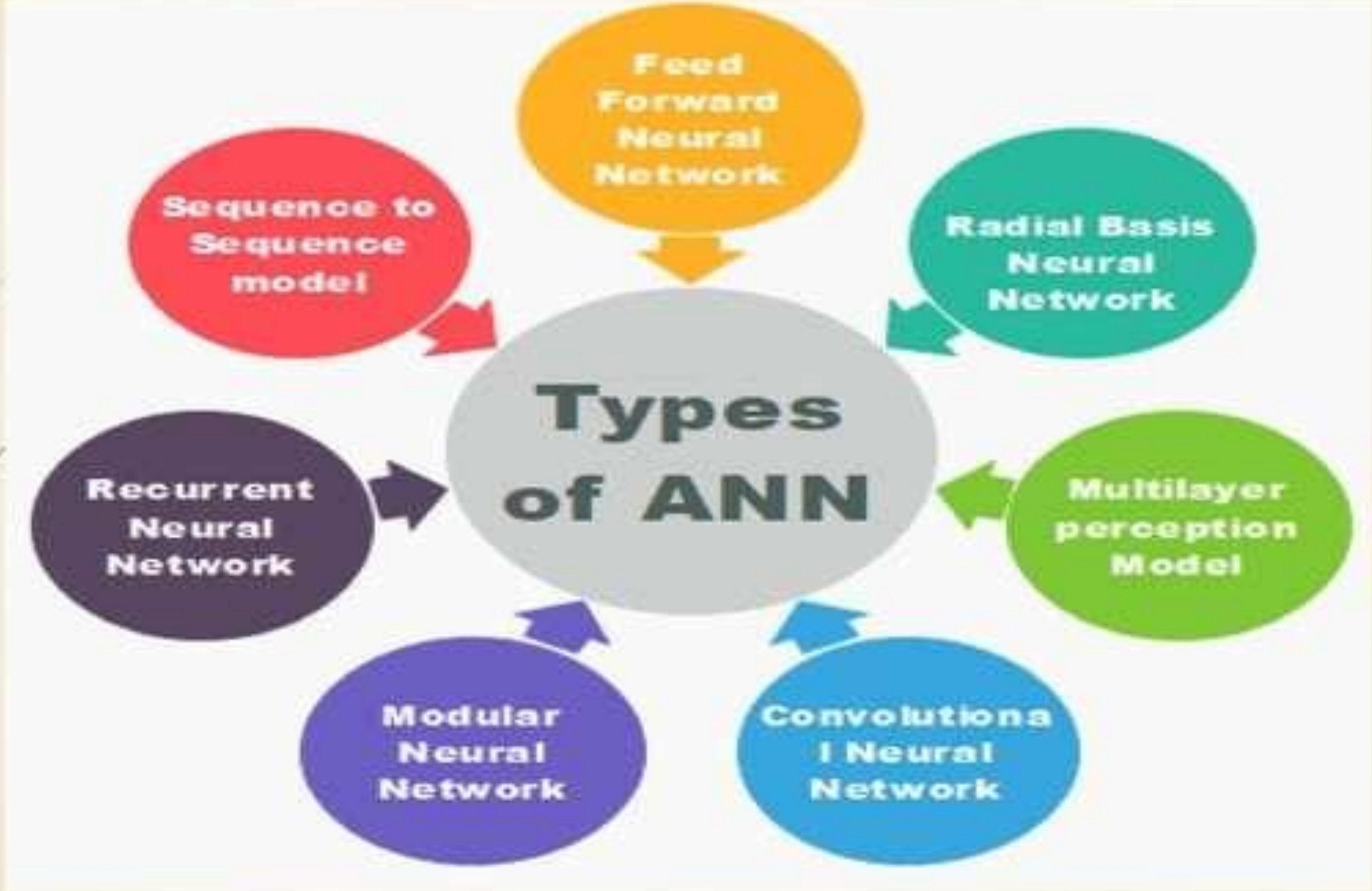
Dendrites
Cell
body
Axon
Axonal arborization
(a)
Bias
x_1
x_2
x_m
w_{k1}
w_{k2}
w_{km}
Inputs
Synaptic weights
(including bias)
Σ
Summing
junction
v_k
Activation
function
$\varphi(\cdot)$
Output
(b)

Process of Neural Network Application





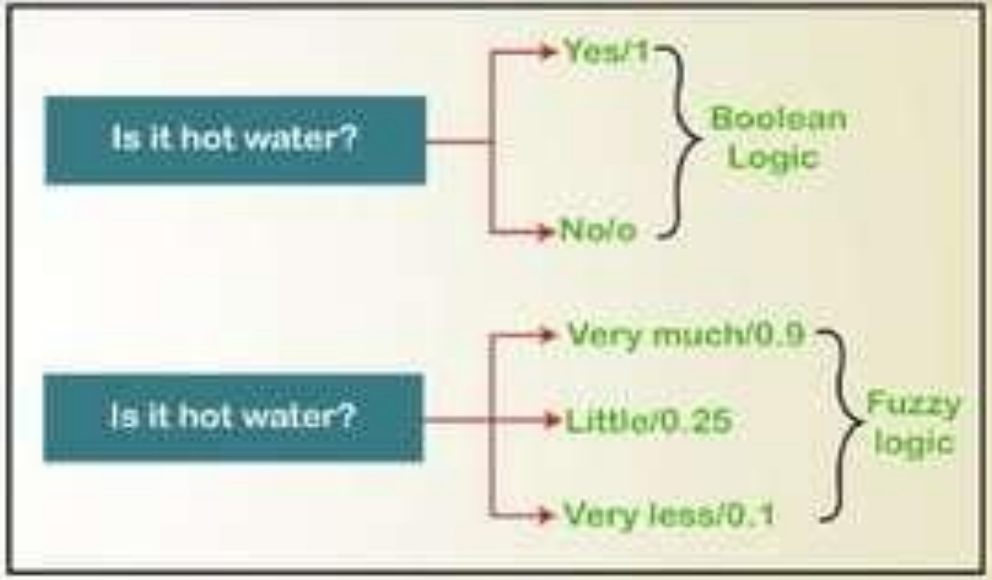

Brain Neuron
Dendrites - It is the input structure
Soma - This is for calculation process
Axon - A channel for the output
https://vinodsblog.com
Artificial Neural Network
Incoming connection - To receive inputs
Activation function – Make a non linear decision
Output connection - Deliver the activation signal
https://AILabPage.com


Feed Forward Neural Network
Sequence to Sequence model
Radial Basis Neural Network
Types of ANN
Recurrent Neural Network
Multilayer perception Model
Modular Neural Network
Convolutiona l Neural Network

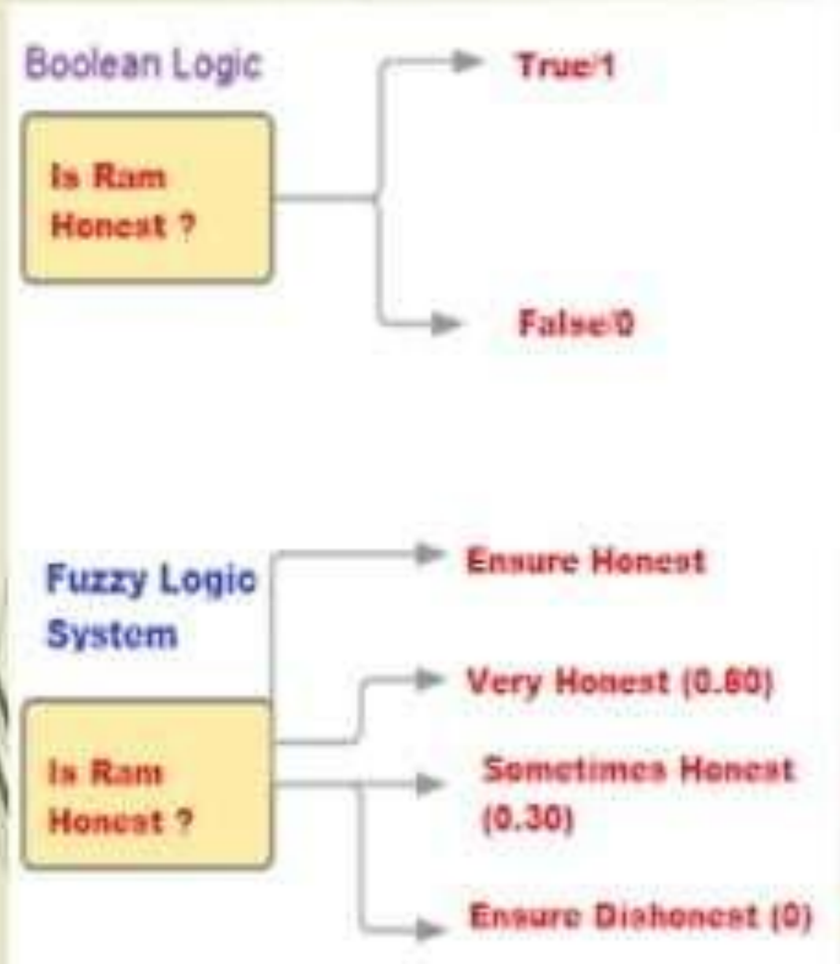
Fuzzy Logic

Fuzzy-"**Not Clear, distinct, or precise; blurred**"

Fuzzy logic is a reasoning method that is **similar to human reasoning**. In other words, a fuzzy logic-based system can make decisions similar to a human.



Fuzzy Logic is a technique that understands the vagueness of a solution and presents the solution with a ***degree of vagueness which is practical to human decision***. It is widely applied in several applications of Artificial Intelligence for reasoning.



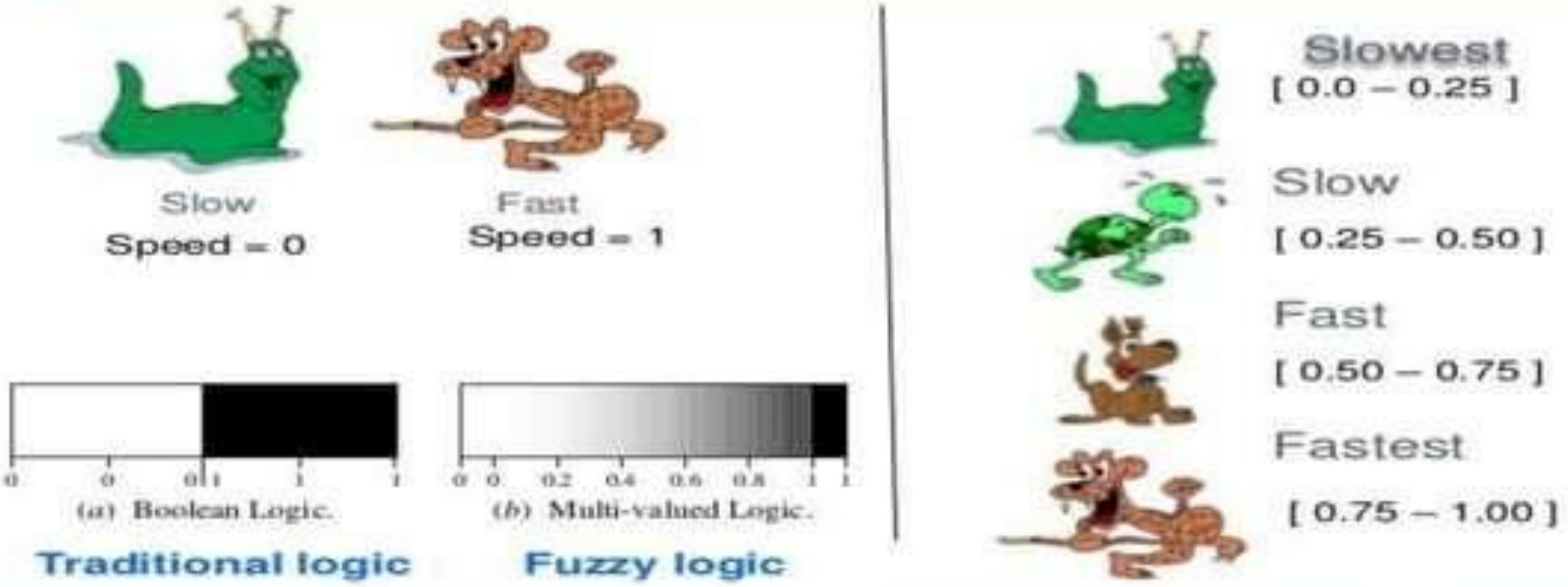
Boolean Logic vs. **Fuzzy Logic**

Isa is 5'10. *vs.* Isa is tall.

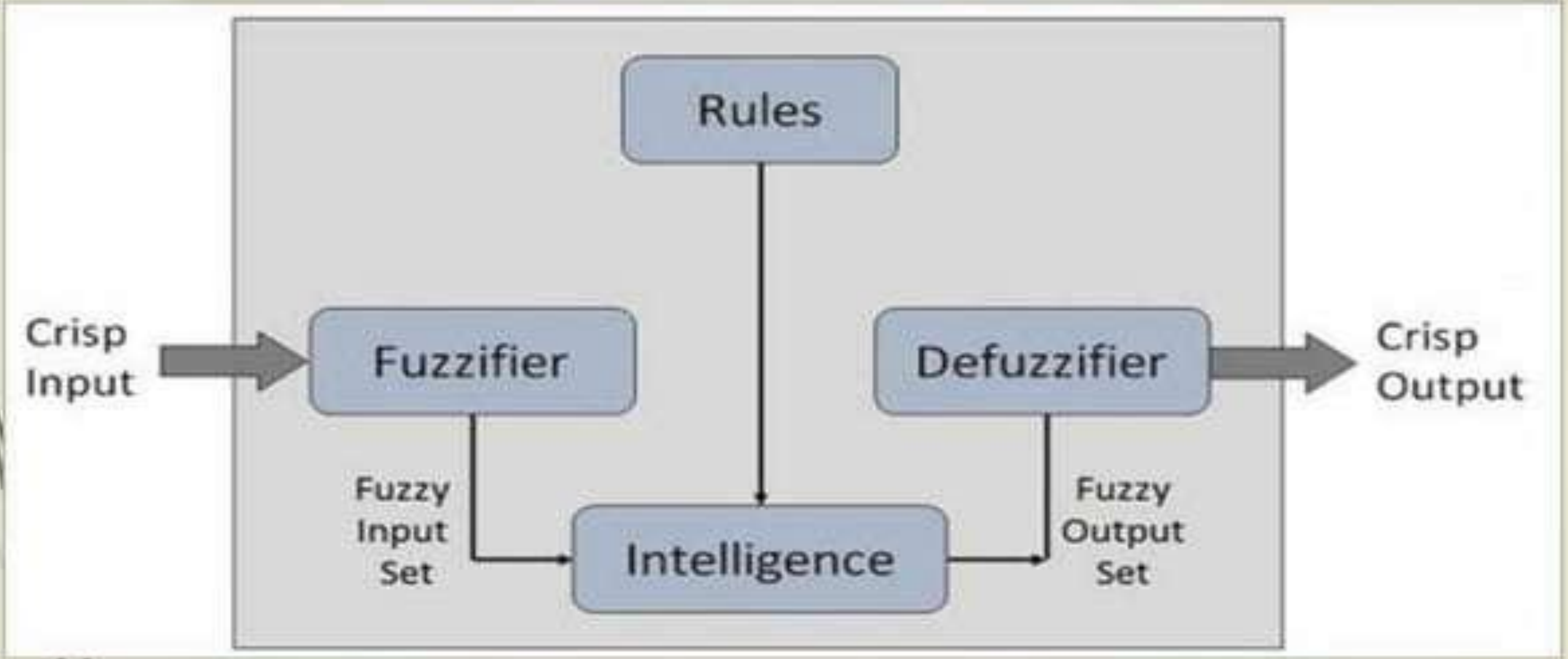
TRUE

Possibly TRUE... but how do we know?

Traditional Logic v/s Fuzzy Logic:



Process of Fuzzy Logic



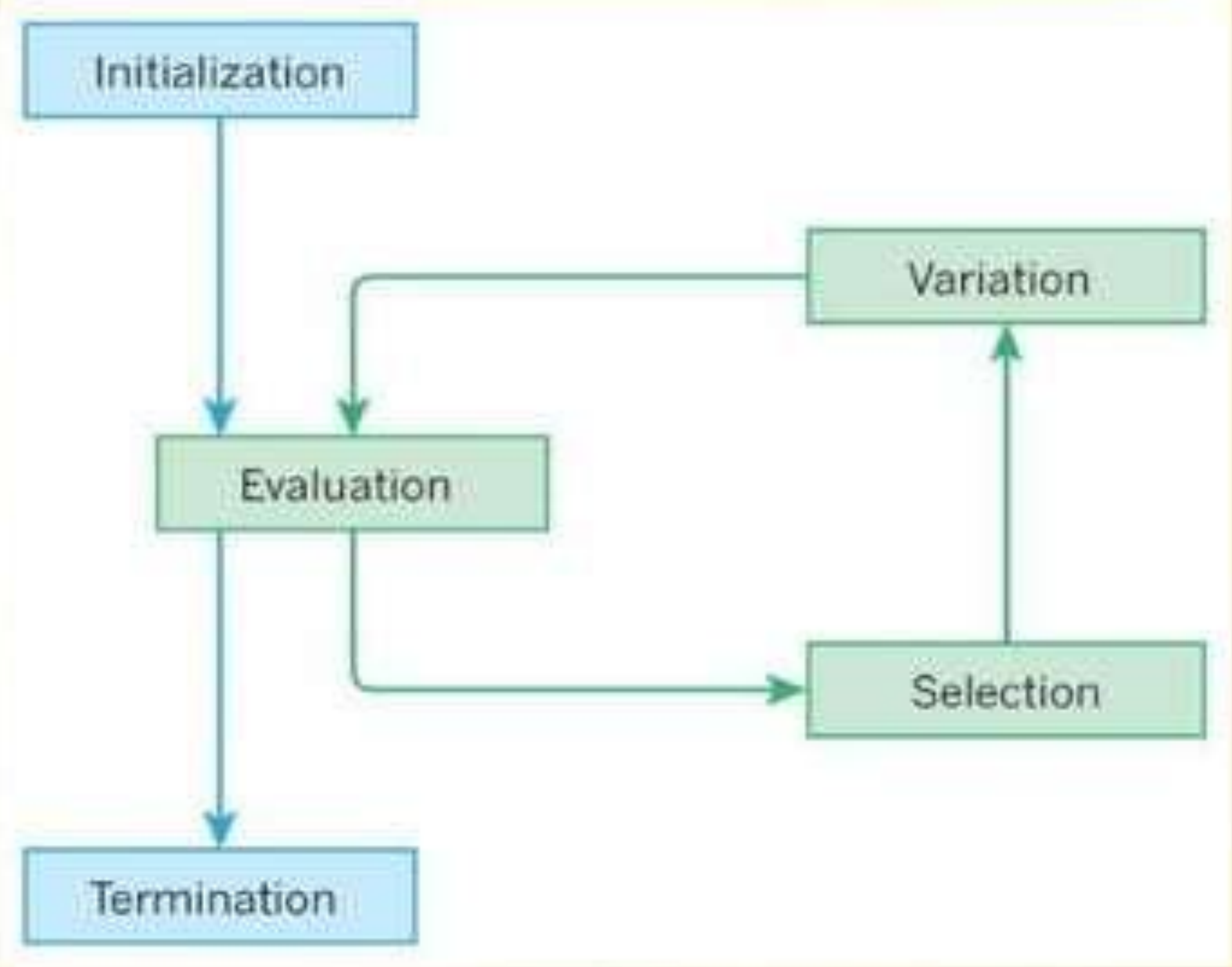
FUZZY LOGIC VERSUS NEURAL NETWORK

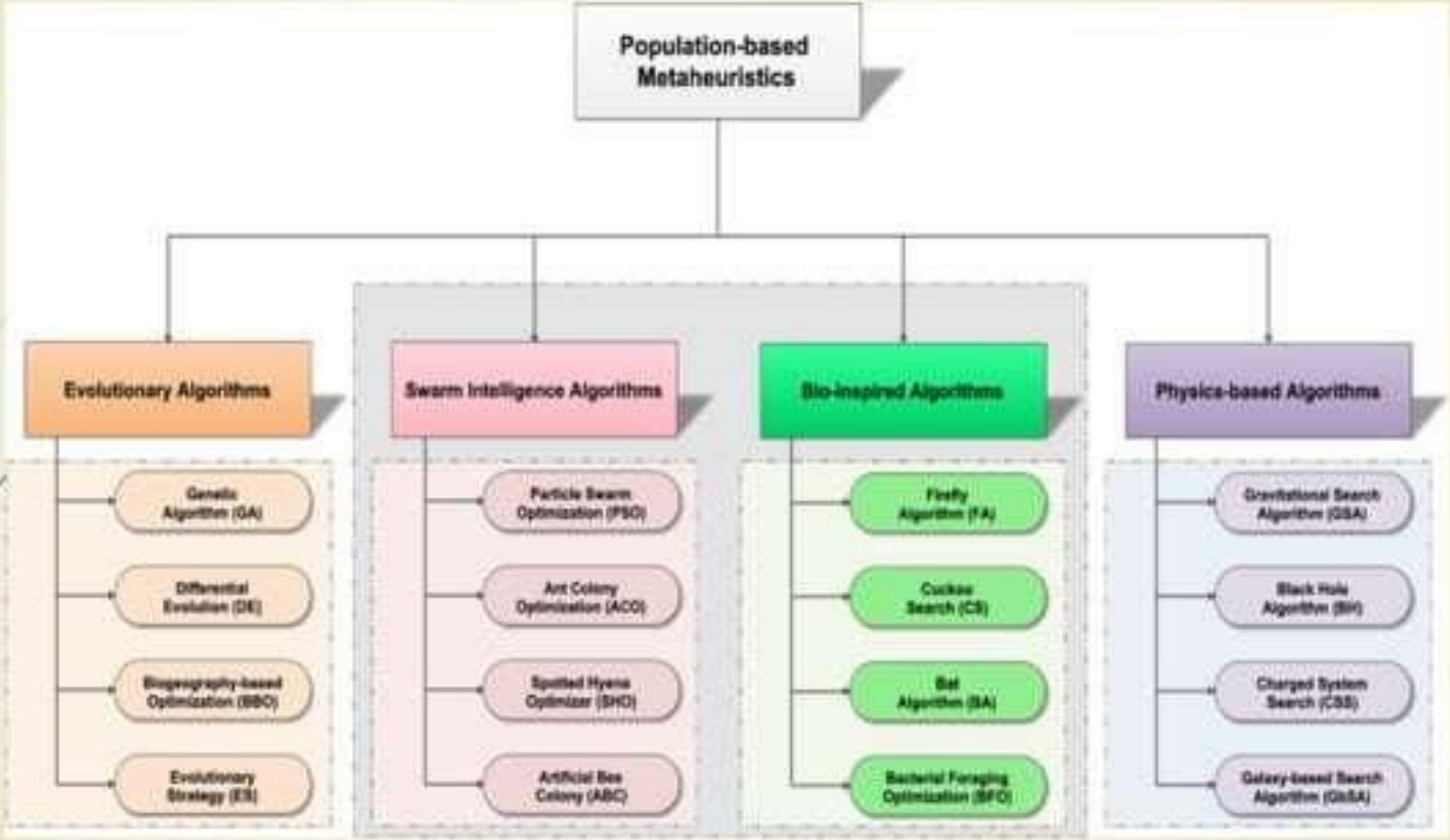
FUZZY LOGIC	NEURAL NETWORK
Reasoning methodology that resembles the human decision making and deals with vague and imprecise information	A system which is inspired by biological neurons in the human brain that can perform computing tasks faster
Helps to perform pattern recognition and classification tasks	Helps to perform prediction, recognition and classification tasks
Simpler than neural network	Complex than fuzzy logic

Visit www.PEDIAA.com

Evolutionary Computation

Evolutionary Computation is a family of optimization algorithms that are ***inspired by biological evolution*** such as Genetic Algorithm, ***survival of creatures*** such as Particle Swarm Intelligence, Ant Colony Optimization, Artificial Bee Colony optimization etc. or any biological processes.




Population-based Metaheuristics
Evolutionary Algorithms
Swarm Intelligence Algorithms
Bio-inspired Algorithms
Physics-based Algorithms
Genetic Algorithm (GA)
Differential Evolution (DE)
Biogeography-based Optimization (BBO)
Evolutionary Strategy (ES)
Particle Swarm Optimization (PSO)
Ant Colony Optimization (ACO)
Spotted Hyena Optimizer (SHO)
Artificial Bee Colony (ABC)
Firefly Algorithm (FA)
Cuckoo Search (CS)
Bat Algorithm (BA)
Bacterial Foraging Optimization (BFO)
Gravitational Search Algorithm (GSA)
Black Hole Algorithm (BH)
Charged System Search (CSS)
Galaxy-based Search Algorithm (GbSA)

Components of Soft Computing

Fuzzy Set Theory	• Uncertainty
Neural Network	• Learning and adaptation
Probabilistic Reasoning	• Reasoning in uncertainty
Evolutionary Computing	• Adaptive search and optimization